Battery End of Life Predictions using Bayesian Hierarchical Modelling

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1 Introduction

The lithium-ion battery is the predominant form of commercial rechargeable batteries, and are wildy used in many applications ranging consumer electronics to electric vehicles and grid storage. Accurately predicting battery end-of-life (EOL), typically defined as the point at which capacity of the battery falls below a specified threshold, is important for ensuring safety, optimizing maintenance schedules, and enabling efficient lifecycle management. However, battery degradation is influenced by a complex interplay of factors such as usage patterns, environmental conditions, and manufacturing variability, leading to substantial heterogeneity across cells and operating groups.

Traditional data-driven approaches to EOL prediction, such as linear regression or machine learning models, often treat all observations as independent and identically distributed. While these models can achieve reasonable predictive performance, they may fail to capture structured variability across groups, such as differences between batches of cells or operating regimes. Recent work has emphasized the importance of incorporating domain structure and uncertainty into predictive models for battery health, particularly in settings with limited data per group or high variability across populations [6].

In this project, we investigate a hierarchical Bayesian model (HBM) for predicting battery EOL that explicitly accounts for group-level heterogeneity following methodology from Zhou et al. [9]. The hierarchical framework allows regression coefficients to vary across groups while sharing statistical strength through common hyperpriors, resulting in a form of partial pooling. This is especially advantageous in battery datasets, where some groups may have few observations but still benefit from information learned across the entire population.

2 Background

The capacity of a lithium-ion battery — the total charge deliverable in a full discharge — fades with repeated cycling. With time and use, storage capacity diminishes and internal resistance increases, driven by degradation mechanisms that can occur simultaneously or trigger one another [3]. A battery is said to reach end-of-life (EoL) when its capacity drops below 80% of its initial value, following the criterion used by Zhou et al. [9].

The rate at which a battery charges or discharges is characterized by the C-rate, defined as the ratio of applied current to nominal capacity [5]. At 1C, a full charge or discharge completes in one hour; at 2C, in thirty minutes. Higher C-rates induce greater mechanical stress during cycling and promote lithium plating during charge [3], both of which accelerate capacity fade. In the dataset used in this study, all cells are discharged at a fixed 4C rate while the charging protocol varies — making the average charging C-rate the primary source of variability in operating conditions across cells.

3 Data Overview

The primary data for this study is sourced from the battery cycling datasets hosted on `data.matr.io` [7].. The consolidated dataset consists of 169 Lithium-ion phosphate (LFP) cells cycled under varying fast-charging conditions. These cells represent 78 distinct charging protocols and have lifetimes that vary accordingly.

3.1 Data Structure and Extraction

The raw data, originally stored in MATLAB format, was ingested into a Python-based workflow following the methodology from Attia et al.'s GitHub [2]. The dataset is organized hierarchically: at the cell level, each battery is indexed by a unique key and carries a charging protocol string and a cycle life — the number of cycles until end-of-life (EoL) is reached. Below this, cycle-by-cycle summary measurements (capacity, internal resistance, temperature) and intra-cycle time-series (voltage, current, capacity curves) are accessible for each cell. A full description of available variables is provided in the Appendix.

3.2 Data Partitioning and Grouping Logic

Zhou and Howey (2023) partitioned the data into eight groups based on their average C-rate and calculated the variance partition coefficient (VPC). Specifically, they calculated:

$$\mathrm{VPC} = \frac{\sigma^2_{\text{group}}}{\sigma^2_{\text{individual}} + \sigma^2_{\text{group}}}$$

$$\sigma^2_{\text{group}} = \mathrm{var}\left(\bar{u}_j - \bar{u}\right), \quad \bar{u} = \frac{1}{N}\sum_{j=1}^{k}\sum_{i=1}^{n_j} y_{ji}$$

$$\sigma^2_{\text{individual}} = \mathrm{var}\left(y_{ji} - \bar{u}_j\right), \quad \bar{u}_j = \frac{1}{n_j}\sum_{i=1}^{n_j} y_{ji}$$

where σ^2_{group} represents lifetime variance between groups and $\sigma^2_{\text{individual}}$ represents the total within-group lifetime variance. They obtained a VPC of 74.5%, implying that between-group variance accounts for the majority of total lifetime variance. It follows that any statistical model which ignores group-level variation would be discarding useful information about the data's variability, while fitting separate statistical models on each group would ignore the remaining within group variability. A hierarchical Bayesian model addresses both by partially pooling estimates across groups.

3.3 Grouping Results

Following Zhou and Howey [9], we first applied constrained K-means clustering on average charging C-rate. While this produced groups with low within-group C-rate variance (see Figure 1), the resulting VPC was only 38.973%. We subsequently grouped cells by their unique charging policy string and optimized the grouping via a hill-climbing algorithm — iteratively reassigning policy strings across the eight groups to maximise VPC — achieving 74.406%. The hill-climb grouping exhibits higher within-group C-rate variance but lower within-group EoL variance (see Figure 2).

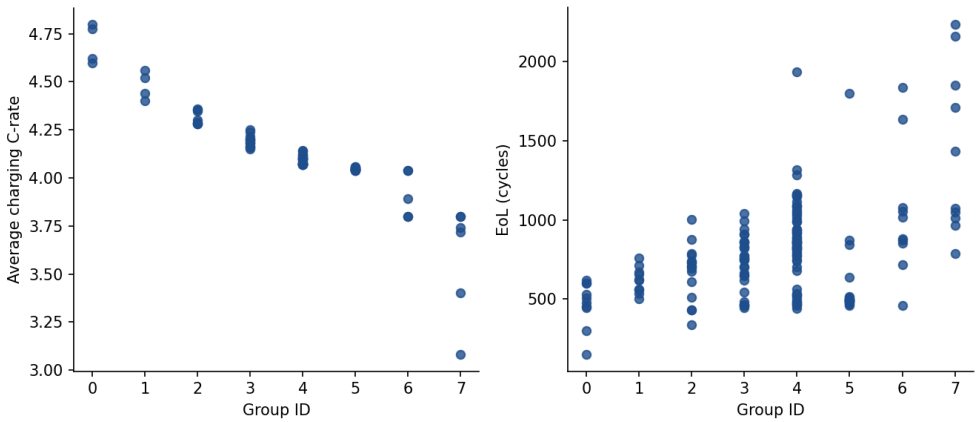

Figure 1: Distribution of average charging C-rate and EoL for each group obtained from constrained K-means.

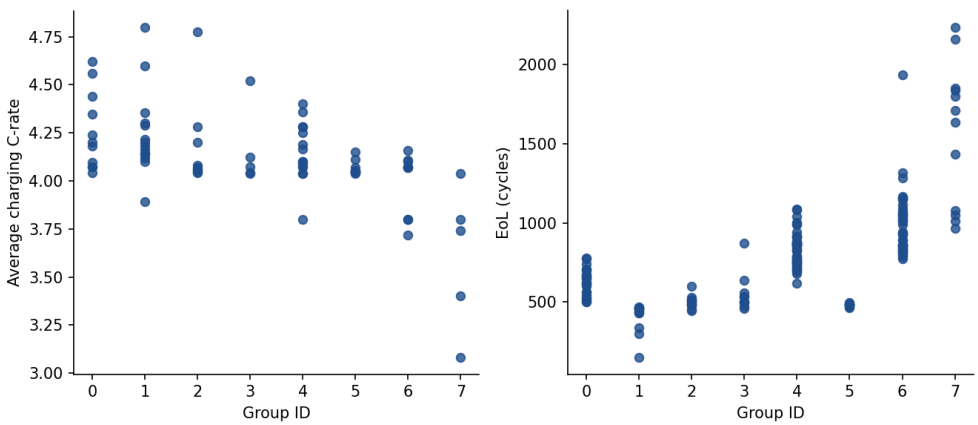

Figure 2: Distribution of average charging C-rate and EoL for each group obtained from hill-climb optimization.

4 Introduction to Hierarchical Bayes

4.1 Multilevel Modelling

To motivate hierarchical modelling, consider first the standard linear regression setting:

$$y = \beta_0 + \beta_1 x_i + \epsilon$$

Now suppose the data naturally contains J groups — for example, homes prices across J states, where the relationship between price and square footage may differ by state. This suggests allowing group-specific parameters:

$$y_i = \alpha_{j[i]} + \beta_{j[i]} x_i + \epsilon_i$$

where j is the group index and $(\alpha_{j[i]}, \beta_{j[i]})$ follow some distribution. For example, they may be jointly normally distributed [4]:

$$\begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_\alpha \\ \mu_\beta \end{pmatrix}, \mathbf{\Sigma} \right), \text{ for } j = 1, \ldots, J$$

4.2 Pooling Tradeoff

In the example above, the multilevel regression may use the same parameters for all groups (complete pooling), or fit entirely separate models for each group with no information shared across them (no pooling). In the latter case, the parameter distribution is specified independently for each group:

$$\begin{pmatrix}\alpha_j \\ \beta_j\end{pmatrix} \sim N\left(\begin{pmatrix}\mu_{\alpha,j} \\ \mu_{\beta,j}\end{pmatrix}, \mathbf{\Sigma}_j\right), \quad j = 1, \ldots, J$$

4.3 Pooling Tradeoff: County Radon Levels

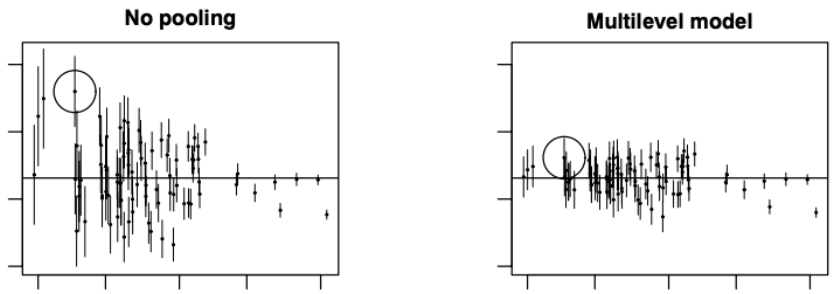

Figure 3: Estimates ± standard errors for average log radon levels in Minnesota counties versus number of observations per county: (a) no-pooling, (b) partial pooling. Counties with fewer observations have more variable estimates and larger standard errors under no pooling, illustrating how small samples produce unreliable county-level estimates.

Gelman et al. [4] analyzed a dataset consisting of radon levels for the 85 counties of Minnesota. The objective was to estimate the distribution of the radon levels of the houses within each county. The first proposed approach was to calculate statistics (e.g., average) using all homes from the sample (complete pooling). This approach does not consider variability among counties and complicates the task of estimating county-specific distributions.

The second proposed approach was to calculate the average log-radon levels in each county (no pooling). With this approach, the authors found that the highest county-specific average log radon level of 85 was much larger than the values from different counties. However, this outlier may not be representative of the actual average log radon levels of its county, since it was obtained from a sample of only two observations. No pooling therefore produces unreliable estimates for data-sparse groups, motivating a middle ground.

4.4 Partial Pooling

Gelman et al. [4] provided a middle ground between complete and no pooling, which they called partial pooling. Specifically, they defined the observation-level model as:

$$y_i \sim N(\alpha_{j[i]} + \beta x_i, \sigma_y^2), \quad i = 1, \ldots, n$$

with the group-level model:

$$\alpha_j \sim N(\mu_\alpha, \sigma_\alpha^2), \quad j = 1, \ldots, J$$

Together they form a two-level structure: in the first equation, each observation is normally distributed around its group's intercept plus a shared slope; in the second, the group intercepts are drawn from a common population distribution.

The two-level structure creates dependency between group-level intercepts through the shared prior $N(\mu_\alpha, \sigma_\alpha^2)$. Specifically, each α_j is estimated from its group's data and also influenced by all other groups. This pulls data-sparse groups toward the population mean μ_α while leaving data-rich groups largely unaffected. The strength of the pull is also determined by σ_α, which is learned from the data.

4.5 Hierarchical Bayes

In the multilevel modelling example above, the parameters μ_α and σ_α^2 are unknown. In Bayesian statistics, these parameters are treated as random variables which are assigned prior distributions. Placing priors on these parameters gives the full hierarchical Bayesian model:

Level 3 (Hyperpriors): $\mu_\alpha \sim p(\mu_\alpha), \quad \sigma_\alpha^2 \sim p(\sigma_\alpha^2)$

Level 2 (Group prior): $\alpha_j \sim N(\mu_\alpha, \sigma_\alpha^2), \quad j = 1, \ldots, J$

Level 1 (Observation model): $y_i \sim N(\alpha_{j[i]} + \beta x_i, \sigma_y^2), \quad i = 1, \ldots, n$

5 Hierarchical Bayesian Model for Battery Lifetime Prediction

5.1 Model Details

Following Zhou & Howey [9], we instantiate the three-level HBM for battery lifetime prediction. The response variable y_{ji} denotes the EoL (in days) for cell i in group j. At the individual level, each cell is represented by a feature vector $\mathbf{x}_{ji} \in \mathbb{R}^4$ constructed from early-cycle measurements; at the group level, each cycling condition group is represented by $\mathbf{g}_j \in \mathbb{R}^2$, which encodes the group's average C-rate. A full description of model variables and features is provided in the Appendix.

Level 3 (Hyperpriors): $\boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, 10\mathbf{I})$

Level 2 (Cycling conditions): $\boldsymbol{\theta}_j \sim \mathcal{N}(\boldsymbol{\gamma}^\top \mathbf{g}_j, \sigma_j^2), \quad j = 1, \ldots, J$

Level 1 (Individual cells): $y_{ji} \sim \mathcal{N}(\boldsymbol{\theta}_j^\top \mathbf{x}_{ji}, \sigma_y^2), \quad i = 1, \ldots, n_j$

The three-level structure encodes two sources of variability simultaneously: individual cell variability within each group (Level 1) and variability in how features relate to EoL across cycling conditions (Level 2). Crucially, at Level 2, the group coefficients $\boldsymbol{\theta}_j$ are not estimated independently — they are drawn from a common distribution centred at $\boldsymbol{\gamma}^\top \mathbf{g}_j$, allowing for information sharing across groups.

5.2 Inference

Inference follows a sequential process. First, the posterior over the hyperparameters is estimated via MCMC [9]:

$$P(\boldsymbol{\gamma} \mid \{Y\}) \propto P(\boldsymbol{\gamma}) \prod_{j=1}^{J} P(Y_j \mid \boldsymbol{\gamma})$$

where $P(\boldsymbol{\gamma}) = \mathcal{N}(\mathbf{0}, 10\mathbf{I})$ is the hyperprior. Since the group parameters $\boldsymbol{\theta}_j$ are unknown at this stage, they are treated as random variables and marginalised out of the likelihood:

$$P(Y_j \mid \boldsymbol{\gamma}) = \int P(Y_j \mid \boldsymbol{\theta}_j) \cdot P(\boldsymbol{\theta}_j \mid \boldsymbol{\gamma})\, d\boldsymbol{\theta}_j$$

The resulting posterior $P(\boldsymbol{\gamma} \mid \{Y\})$ has no closed form due to the intractable normalising constant $P(\{Y\})$; MCMC is therefore used to obtain samples from it. These samples are then used as a prior to obtain the group-level posteriors $P(\boldsymbol{\theta}_j \mid \boldsymbol{\gamma}, Y_j)$ via Bayes' rule [9]. Starting from the joint distribution, we have:

$$P(\boldsymbol{\theta}_j, \gamma, Y_j) = P(Y_j \mid \boldsymbol{\theta}_j) \cdot P(\boldsymbol{\theta}_j \mid \gamma) \cdot P(\gamma)$$

Conditioning on γ and Y_j and normalizing gives:

$$P(\boldsymbol{\theta}_j \mid \gamma, Y_j) = \frac{P(Y_j,\, \boldsymbol{\theta}_j,\, \gamma)}{P(Y_j \mid \gamma)} \tag{1}$$

$$= \frac{P(Y_j \mid \boldsymbol{\theta}_j,\, \gamma) \cdot P(\boldsymbol{\theta}_j,\, \gamma)}{P(Y_j \mid \gamma)} \tag{2}$$

$$= \frac{P(Y_j \mid \boldsymbol{\theta}_j) \cdot P(\boldsymbol{\theta}_j \mid \gamma) \cdot P(\gamma)}{P(Y_j \mid \gamma)} \tag{3}$$

where in the third line we use the fact that once $\boldsymbol{\theta}_j$ is known, γ carries no additional information about Y_j, so $P(Y_j \mid \boldsymbol{\theta}_j,\, \gamma) = P(Y_j \mid \boldsymbol{\theta}_j)$.

6 Experiment

The analysis was completed in Python, and the hierarchical Bayesian model was built in Python using the PyMC package [1].

First, the data set was processed following section 3. Early cycle features described in table 9 were extracted and the data was partitioned into 8 groups as described in section 3.3. The following experiment uses the hill-climbing grouping, which yielded a higher VPC (74.5% vs. 39.0% for K-means) — indicating greater between-group variance relative to total variance (section **??**), and thus a more statistically justified hierarchical structure; the HBM under K-means grouping confirmed this, showing no improvement over the baseline. The data was split into training and test sets of 135 and 34 cells respectively, stratified by cycling group g. A 5-fold cross-validation was performed 4 times (yielding 20 error estimates) to assess the generalizability of the model.

7 Results and Discussion

The performance of the Hierarchical Bayesian model is compared against two baselines: a ridge regression model with just the features in Table 9, and one including the cycling group g as well. Root mean error squared (RMSE) and mean absolute percentage error (MAPE) are used as metrics to evaluate the performance of the models. We consider MAPE as an additional metric because

RMSE is significantly influenced by outliers.

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{N}}$$

$$\text{MAPE} = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

	Baseline (without g)	Baseline (with g)	HBM
RMSE	9.31	6.51	7.23
MAPE	22.5%	16.8%	14.7%

Table 1: Average RMSE (in days) and MAPE of the EoL predictions on the cross validation

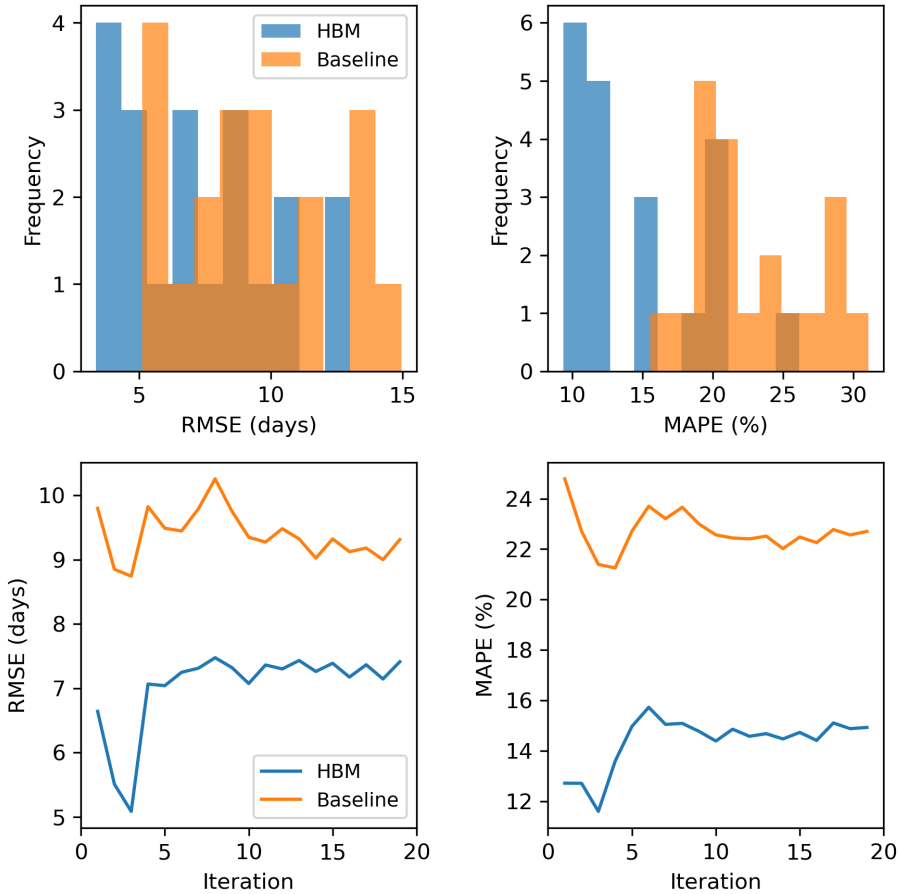

Figure 4: Histogram of all RSME and MAPE results for the HBM and baseline

Table 1 shows the average RMSE and MAPE after each repetition and fold, and figure 4 shows the RMSE and MAPE values for each fold and repetition. HBM outperforms the baseline by 28.7% with RMSE, but has similar results to the baseline with g as a feature. This indicates the importance of the cycling condition as a feature and motivates the consideration of a hierarchical

model, since conventional regression would treat the cycling condition and other features the same. Additionally, HBM outperforms both baseline models when considering MAPE.

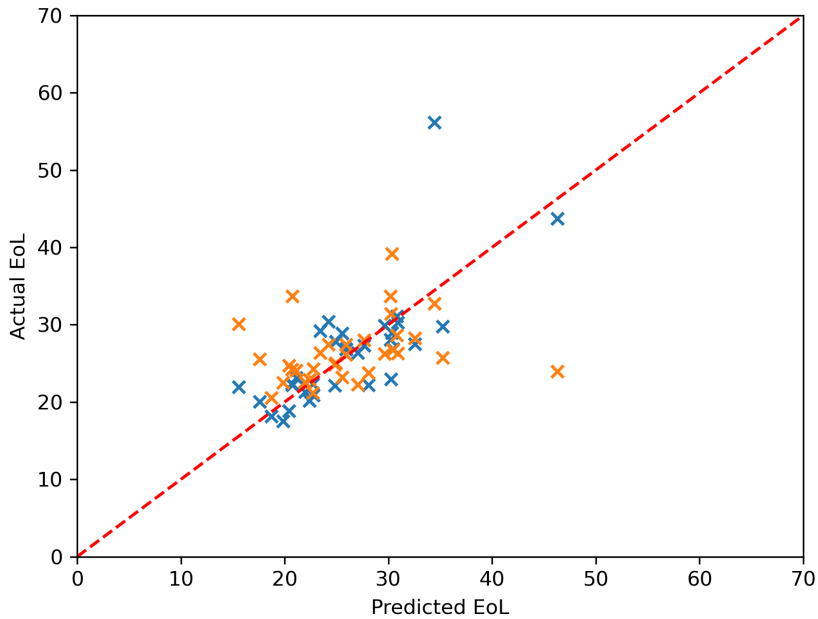

Figure 5: Predicted v.s. actual EoL for HBM and baseline

	Baseline (without g)	Baseline (with g)	HBM
RMSE	18.2	18.2	18.8
MAPE	18.5%	17.1%	13.9%

Table 2: RMSE (in days) and MAPE of the EoL predictions on the test set

Now using just a training and test set, table 2 shows the RMSE and MAPE on the test set predictions and figure 5 shows the EoL predictions from the HBM and baseline models compared to the true values. On the test set, HBM achieves a MAPE of 13.9%, outperforming both baselines, though its RMSE of 18.8 days is marginally higher than the baselines (18.2 days). The divergence between RMSE and MAPE on the test set suggests the presence of outliers that disproportionately inflate the HBM's squared error, while the model generalizes better on the majority of cells. This is consistent with the cross-validation results, where HBM's advantage was clearest under MAPE.

To assess model validity, table 3 shows MCMC convergence diagnostics for the final model fit. All parameters achieved $\hat{R} \leq 1.01$ and bulk effective sample sizes above 894, indicating that the chains mixed well and the posterior estimates are reliable [8]. The observation noise parameter σ_y was tightly estimated at 0.55 ± 0.04 (standardized units) with an ESS of 3433, showing high posterior certainty about the residual variance.

The hyperprior slope $\gamma_{\text{slope},0}$, governing how mean C-rate modulates the group-level intercept, has a posterior mean of -3.27 with a 94% HDI of $[-4.91, -1.55]$, which excludes zero. This provides credible evidence that groups with higher average charge rates are associated with shorter battery lifetimes at the group level, and justifies the hierarchical structure of the model. Furthermore,

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	$\hat{R}$
gamma_intercept[0]	13.405	3.616	6.303	20.183	0.109	0.106	1148.00	1676.00	1.00
gamma_intercept[1]	2.586	1.526	-0.247	5.346	0.041	0.058	1558.00	1814.00	1.00
gamma_intercept[2]	-0.047	1.706	-3.254	3.168	0.038	0.028	2012.00	2111.00	1.00
gamma_intercept[3]	1.128	2.635	-4.206	5.756	0.067	0.052	1561.00	1997.00	1.01
gamma_slope[0]	-3.269	0.873	-4.913	-1.545	0.026	0.026	1150.00	1656.00	1.00
gamma_slope[1]	-0.601	0.371	-1.266	0.10	0.01	0.014	1545.00	1773.00	1.00
gamma_slope[2]	0.01	0.409	-0.749	0.793	0.009	0.007	2002.00	2135.00	1.00
gamma_slope[3]	-0.261	0.628	-1.475	0.901	0.016	0.012	1564.00	1985.00	1.01
sigma_theta[0]	0.472	0.226	0.134	0.935	0.007	0.006	894.00	897.00	1.00
sigma_theta[1]	0.097	0.091	0.00	0.256	0.002	0.003	1725.00	1773.00	1.00
sigma_theta[2]	0.086	0.076	0.00	0.216	0.002	0.002	1797.00	1840.00	1.00
sigma_theta[3]	0.115	0.116	0.00	0.307	0.003	0.006	1485.00	1467.00	1.00
sigma_y	0.55	0.037	0.481	0.615	0.001	0.001	3433.00	3011.00	1.00

Table 3: MCMC summary statistics for the final model parameters

$\sigma_{\theta,0} = 0.47$ with HDI $[0.13, 0.94]$ confirms that meaningful variation in baseline lifetime persists across groups beyond what C-rate alone explains. In contrast, the hyperprior slopes for the early-cycle features $\gamma_{\text{slope},1-3}$ all have HDIs that include zero, indicating that the regression coefficients for F1, F2, and F3 are shared across groups rather than modulated by cycling condition. The hierarchy therefore operates primarily through group intercepts, while feature effects generalize across charging protocols.

8 Conclusion

This work investigated the use of a hierarchical Bayesian model (HBM) for predicting battery end-of-life (EoL) from early-cycle features, incorporating cycling condition as a group-level structure rather than a flat input feature. Three early-cycle features were extracted following the approach of [6], and cells were partitioned into 8 groups based on charging protocol using a hill-climbing algorithm, which produced a variance partition coefficient sufficient to justify a hierarchical treatment of the data.

Across 20 cross-validation folds, the HBM achieved a mean MAPE of 14.7%, outperforming both the plain ridge baseline (22.5%) and the ridge model with the group feature (16.8%). On the held-out test set, the HBM maintained the lowest MAPE at 13.9%, though its RMSE was marginally higher than the baselines, suggesting sensitivity to a small number of outlier predictions. MCMC diagnostics confirmed reliable posterior inference, with all parameters satisfying $\hat{R} \leq 1.01$ and adequate effective sample sizes.

Bayesian posterior analysis revealed that the hyperprior slope linking mean C-rate to the group intercept was credibly negative (HDI $[-4.91, -1.55]$), providing probabilistic evidence that higher charge rates are associated with reduced battery lifetime at the group level. The early-cycle feature coefficients, by contrast, were found to generalize across groups, with no credible C-rate modulation. Together, these findings suggest that the performance advantage of the HBM over conventional regression stems not from learning different feature sensitivities per group, but from appropriately pooling information across cells sharing similar cycling conditions when estimating baseline lifetime.

Future work could explore richer group structures and the influence of cluster size and number. The grouping of the cycling conditions heavily influenced the results of HBM as well as the baseline rigde regression with the group feature.

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Appendix

Data Dictionary

Table 4: Cell Level Variables

Variable	Description	Type
`cycle_life`	Number of cycles until EoL	Float
`charge_policy`	Charging protocol	String
`summary`	Metadata for each cycle	Dictionary
`cycles`	Time series of measurements	Dictionary

Table 5: Summary Level (Cycle-by-Cycle Metadata)

Variable	Description	Type
`IR`	Internal resistance	Numeric array
`QC`	Charge capacity	Numeric array
`QD`	Discharge capacity	Numeric array
`Tavg`	Average temperature	Numeric array
`Tmin`	Minimum temperature	Numeric array
`Tmax`	Maximum temperature	Numeric array
`chargetime`	Charge time	Numeric array
`cycle`	Cycle number	Numeric array

Table 6: Cycle Level (Intra-cycle Time Series)

Variable	Description	Type
`I`	Current	Numeric array
`Qc`	Charge capacity	Numeric array
`Qd`	Discharge capacity	Numeric array
`Qdlin`	Linear interpolated discharge capacity	Numeric array
`T`	Temperature	Numeric array
`Tdlin`	Linear interpolated temperature	Numeric array
`V`	Voltage	Numeric array
`dQdV`	Derived vectors of discharge	Numeric array
`t`	Time	Numeric array

Model Variables and Features

Table 7: Variables

Symbol	Description
$j = 1, \ldots, J$	Cycling condition group index
$i = 1, \ldots, n_j$	Cell index within group j
y_{ji}	EoL time (days) for cell i in group j
$\boldsymbol{\theta}_j \in \mathbb{R}^4$	Regression coefficients for group j
σ_j^2	Variance of group j model
$\boldsymbol{\gamma} \in \mathbb{R}^{4\times 2}$	Hyperparameter matrix linking cycling conditions to group coefficients

Table 8: Feature vectors

Symbol	Components	Level
$\mathbf{x}_{ji} \in \mathbb{R}^4$	$(1, F1_{ji}, F2_{ji}, F3_{ji})^\top$	Individual
$\mathbf{g}_j \in \mathbb{R}^2$	$(1, \bar{g}_j)^\top$	Group

Table 9: Individual-level features (computed from first 100 cycles)

Symbol	Description
$F1_{ji}$	Variance of $\Delta Q(V)$ between cycles 10 and 100
$F2_{ji}$	Minimum of $\Delta Q(V)$ over the same interval
$F3_{ji}$	Discharge capacity at cycle 2

[1]

[1] $\Delta Q(V)$ denotes the discharge curve difference as a function of voltage, following Severson et al. (2019) [6].